Aviroop Pal

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About

Enthusiastic Machine Learning and Backend Developer with a foundation in building scalable backend systems, deep learning models, and containerized applications. Skilled in Python, PyTorch, TensorFlow, and modern web frameworks (FastAPI, Django, Flask, React). Research-driven, with hands-on implementations of advanced architectures such as Transformers, ResNet, and Federated Learning. Experienced in DevOps/MLOps practices including Docker, CI/CD pipelines, and production-level deployment. Passionate about applying AI, NLP, and Computer Vision to solve real-world problems with a focus on scalability, privacy, and interpretability.

Technical Skills

- **Programming & Development**: Exposure with Python, Go (Golang), C, JavaScript, FastAPI, Django, Flask, React, Node.js, Express.js,Next.js, Streamlit, Docker, Git, GitHub Actions (CI/CD pipelines), WSL, SQLAlchemy (SQLite, etc.), Pydantic, Loguru, Postman, and Thunder Client.
- **Backend Development**: Extensive experience in building and deploying scalable backend applications using Flask, Django, and FastAPI. Proficient in designing RESTful APIs, integrating with databases (SQLAlchemy, SQLite), and ensuring seamless communication between the frontend and backend services.
- **Machine Learning & AI**: Practitioner for Machine Learning, Deep Learning, TensorFlow, PyTorch, Keras, scikit-learn, CNN, LSTM, YOLO, BERT, Mistral, LoRA, Transformers, LangChain, Retrieval-Augmented Generation (RAG), and model serialization/deserialization using Pickle.
- **Data Science & Analysis**: Experience with Data Analysis, Data Visualization, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, and applying models to real-world problems such as footfall analysis and vision-specific projects.
- **Natural Language Processing (NLP)**: Practitioner in NLP frameworks such as NLTK, Transformers, BERT, and LangChain for building conversational AI systems and RAG-based applications.
- **DevOps ,MLOps and CI/CD**: Experience with Docker, Git, GitHub Actions for CI/CD pipelines, and production-level libraries like Pydantic, Loguru, and Poetry for robust application development.
- **Computer Vision**: Competent in OpenCV, CNN, YOLO architectures for image recognition and vision-specific projects.
- **Communication & Collaboration**: Strong in Effective Communication, Collaboration and Team Building.
- **Languages**: Fluent in English , Hindi and Bengali with full proficiency.

Experience

AI/ML Intern *(Jan 2025 – Feb 2025)*

- **Organization: TherapyU | Location: Birac, Adamas University, West Bengal | Mode: Hybrid | Duration: 1 month**
- Developed and implemented AI/ML algorithms to enhance the functionality of a web application.
- Collaborated with UI/UX and development teams to integrate AI features seamlessly.
- Conducted data analysis, preprocessing, and model optimization for performance and scalability.
- Assisted in designing innovative AI-based solutions and provided technical documentation.
- **Technologies**: Python, Machine Learning, AI Integration, Data Analysis, Web Application Development

Software Development Intern

- **Organization: Proption Fintech Private Limited | Location: Delhi | Mode: Remote | Duration: 4 month**
- Developed internal company tools using Go(Golang) to streamline operations.
- Designed and implemented financial algorithms, including the Supertrend indicator for NIFTY, supporting data-driven decision-making.
- Contributed to both backend and frontend development, assisting in feature implementation and system enhancements.
- Worked on data collection, preprocessing, and analysis to support algorithm development and performance evaluation..
- **Technologies:** Go (Golang), Backend Development, Frontend Development, Financial Algorithms, Data Analysis

AI/ML Intern

- **Organization: BaseApp | Location: Delhi| Mode: Remote| Duration: 2 month**
- Developed a real-time Voice-to-Voice (V2V) AI system with sub-second latency for conversational AI applications.
- Built an end-to-end pipeline integrating Speech-to-Text , Large Language Models , and Text-to-Speech.
- Implemented a WebSocket-based streaming API for bidirectional audio communication and real-time response generation.
- Designed a scalable asynchronous backend architecture with Redis-based session management and Dockerized deployment.
- Ensured production readiness through authentication, DoS protection, memory optimization, and GPU acceleration.
- **Technologies:** Python 3.11, Faster-Whisper, LiteLLM, Ollama, Piper TTS, WebSockets, Redis, Docker, REST APIs, Real-Time Systems, Machine Learning

Projects and Research Experience

Freelance/Academic Projects
07/2023 – Present

- **Research Paper Implementations**
  Implemented key architectures from research papers, such as **"Attention Is All You Need"** for transformer-based models and **"Deep Residual Learning for Image Recognition"** (ResNet) for enhanced image classification tasks.
  Technologies: Transformers, PyTorch, TensorFlow
  Impact: Achieved improved accuracy in text summarization and image recognition tasks by applying state-of-the-art research methods.
- **Ongoing Research on Federated Learning and Differential Privacy**
  Currently working on a project implementing concepts from **Federated Learning** research, focusing on privacy-preserving machine learning models that allow decentralized data processing across multiple nodes.
  Technologies: Federated Learning Frameworks,DP, TensorFlow, PyTorch
  Goal: To develop a scalable federated learning system that ensures data privacy while maintaining model performance across distributed environments.

Projects

- **Character-level Language Modeling** *(Jul 2024 – Jul 2024)*
  - Developed a compact Transformer model (0.209729M parameters) for character-level language modeling.
  - Designed to learn text patterns and generate new text mimicking the style of training data.
  - **Technologies:** Transformers, PyTorch, Python
- **Advanced Multi-Modal Crop Health Classification System**
  - Built a deep learning system in PyTorch that fuses multi-spectral imagery, temporal sensor data, and environmental context for crop health classification.
  - Implemented ResNet-style CNN with attention, LSTM for temporal analysis, and cross-modal attention for data fusion.
  - Added uncertainty estimation and advanced training strategies (Focal Loss, AdamW, Cosine Annealing) for reliable and interpretable predictions.
  - **Technologies:** PyTorch, ResNet, LSTM, Attention Mechanisms, Computer Vision, Time Series Analysis, Machine Learning
- **Cancer Digital Twin**
  - Developed a computational framework that creates a virtual model of a patient's cancer by integrating clinical, genomic, and imaging data.
  - Implemented risk assessment, treatment simulation, and disease progression modeling using Markov processes and survival analysis.
  - Built a modular FastAPI backend for scalable and personalized cancer predictions.
  - **Technologies:** Python, FastAPI, Survival Analysis, Markov Models, Machine Learning, Clinical Data Integration
- **Python Library (Docker Automation)** *(Jun 2024 – Aug 2024)*
  - Created a Python library that automatically Dockerizes a single-model machine learning application with a single command-line instruction.
  - **Technologies:** Python, Docker, FastAPI
- **dock-slimscheck** *(v1.0)*
  - Developed a comprehensive **Dockerfile analyzer** that combines static analysis with optimization and security recommendations—like having a senior DevOps engineer reviewing your Dockerfile.
  - Provides context-aware suggestions for base image optimization, layer efficiency, best practices, and security scanning, with detailed, actionable fixes and explanations.
  - **Technologies:** Python, Docker, DevOps Tooling, Static Analysis, Security Scanning

Education

Bachelor in Computer Science (Under-Graduate-3rd Year) (Major: Computer Science and Engineering.**) CGPA: 8** **Adamas University**

Certification

- **Generative AI Fundamentals**, Google.
- **Prepare Data for ML APIs on Google Cloud**, DeepLearning.ai.
- **Create and Manage cloud resources**, Google.
- **Enterprise Design Thinking** , IBM.
- **Build a Secure Google Cloud Network**, Google